

ANA 252 — General Embryology & Neuroanatomy

Practice Workbook & Answer Key | ABUAD 200 Level | Second Semester

An original revision aid by the EverythingABUAD student team | everythingabuad.com

How to use this workbook. The questions below are grouped by topic and were written from scratch by our team to help you revise the core concepts of ANA 252 (general embryology and neuroanatomy). They are original practice items for self-testing — not past exam papers and not a prediction of any future test. Try to answer each question on your own first, then check yourself against the answer key at the end. Always cross-check with your lecturer's current course outline.

Fertilization & the Acrosome

1. The acrosome is a cap-like vesicle over the sperm head. If a point mutation altered the enzymes stored inside it, which step of fertilization would fail, and why?

Implantation & the Placenta

2. In placenta accreta the placenta anchors abnormally deep into the myometrium. Which cell type mediates this attachment, and at what developmental stage does it first appear?

The Female Reproductive System

3. List the internal organs of the female reproductive system.
4. Over roughly what interval do the cyclic changes of the ovary and uterus recur?
5. Roughly between which ages does a woman's fertile (reproductive) span run?
6. In which organ does the menstrual (uterine) cycle physically take place?

The Ovarian Cycle

7. How many phases make up the ovarian cycle, and name them.
8. At approximately what point in the cycle does ovulation occur?
9. Which half of the cycle contains the follicular phase, and which contains the luteal phase?
10. By what other names is the follicular phase known?

The Sequence of Fertilization

11. Arrange these fertilization events in their correct order: acrosomal digestion of the zona pellucida; penetration of the corona radiata; the cortical reaction; binding to the oolemma (vitelline membrane).

Gametogenesis (Oogenesis & Spermatogenesis)

- 12. A primary oocyte is arrested in prophase I from before birth until puberty. What is the DNA content - chromosomes and chromatids - of this cell during childhood?
- 13. How do oogenesis and spermatogenesis differ in the number of functional gametes and polar bodies produced from one precursor cell?
- 14. Which of the two processes produces the larger gamete?
- 15. Approximately how long does the complete process of spermatogenesis take?

Ovulation

- 16. At ovulation, which cell is expelled, and from which structure is it released?

Sperm Production & Storage

- 17. Where are spermatozoa produced, and where are they temporarily stored before ejaculation?

The Preovulatory (Follicular) Phase

- 18. Besides maturing the ovarian follicle, what key hormonal event occurs during the preovulatory phase?
- 19. Which follicle secretes the estrogen of the preovulatory phase?
- 20. Which phase of the uterine cycle does this estrogen drive?

The Second Week of Development

- 21. Which major event of implantation is completed during the second week?
- 22. Describe the hypoblast layer and its position early in the second week.
- 23. What forms in the trophoblast around day 9, and what do these spaces become?
- 24. Which two structures form the lining of the exocoelomic (primitive yolk sac) cavity?

The Embryonic Germ Layers

- 25. Which germ layer gives rise to the central and peripheral nervous systems?
- 26. Name organ systems derived from the mesoderm.
- 27. Which germ layer forms the epithelial lining of the respiratory and digestive (alimentary) tracts?

Hormonal Control of the Ovarian Cycle

- 28. Which pituitary hormone drives a primary follicle to mature into a Graafian follicle?
- 29. What is the usual feedback effect of estrogen on FSH secretion?
- 30. After ovulation, which hormone converts the corpus hemorrhagicum into the corpus luteum?

Capacitation

- 31. What is capacitation, and where does it take place?
- 32. What happens to sperm movement during capacitation?
- 33. Why is capacitation essential before fertilization?

The Zona Pellucida

- 34. What is the zona pellucida, and what is its main chemical composition?
- 35. Which zona glycoprotein acts as the primary sperm receptor?

Organization of the Nervous System

- 36. What are the components of the central nervous system (CNS)?
- 37. What does the peripheral nervous system (PNS) include?
- 38. Does the somatic nervous system control voluntary or involuntary functions?
- 39. What are the protective membranes that cushion the CNS called?

Neuroglia

- 40. What are neuroglia?
- 41. Which glial cells help form the blood-brain barrier?
- 42. Which part of the neuron is responsible for propagating the signal?
- 43. Which glial cells line the ventricles and the central canal?

The Hindbrain

- 44. Which of these belong to the hindbrain: cerebellum, medulla oblongata, pons, cerebrum?

Synaptic Transmission

- 45. From which structure are neurotransmitters released?
- 46. Into what space are neurotransmitters released, and how wide is it?
- 47. Which part of the neuron integrates incoming signals?

The Autonomic Nervous System

- 48. Which autonomic division mediates the 'fight-or-flight' response?
- 49. Are autonomic functions voluntary or involuntary?
- 50. Which neurotransmitter do most postganglionic sympathetic fibres release, and what is the main exception?
- 51. Is the enteric nervous system regarded as part of the ANS?

The Cerebral Cortex

- 52. Which lobe processes somatic (general body) sensation?
- 53. Which lobe controls voluntary motor activity?
- 54. Which lobe is the primary centre for vision, and which for hearing?

The Spinal Cord

- 55. Through which opening is the brain continuous with the spinal cord?
- 56. Through which opening do spinal nerves leave the vertebral column?
- 57. How many pairs of spinal nerves are there?
- 58. At what vertebral level does the spinal cord end in an adult?

The Blood-Brain Barrier

- 59. What structure primarily forms the blood-brain barrier?
- 60. Which additional cells contribute to blood-brain-barrier function?
- 61. Is the blood-brain barrier fully mature at birth?
- 62. Which cross the barrier more readily, lipid-soluble or water-soluble substances?

Circumventricular Organs (CVOs)

- 63. What distinguishes circumventricular organs from the rest of the brain?
- 64. Name a CVO involved in the vomiting reflex.
- 65. Where in the brain are the CVOs located?

Transport Across the BBB

- 66. How does glucose cross the blood-brain barrier?
- 67. How do large neutral amino acids cross the barrier?
- 68. Must every CNS-active drug be lipid-soluble?

BBB Breakdown in Disease

- 69. Name conditions that can disrupt the blood-brain barrier.
- 70. Is blood-brain-barrier disruption in stroke beneficial or harmful?
- 71. Why does edema develop after a traumatic brain injury?

Cerebrospinal Fluid (CSF) Production

- 72. Where is most cerebrospinal fluid produced?
- 73. Approximately how much CSF is produced each day, and what is the total volume present at any time?

74. Does CSF production fall as intracranial pressure rises?

CSF Circulation

75. Through which opening does CSF pass from the lateral ventricles to the third ventricle?

76. Through which openings does CSF leave the fourth ventricle into the subarachnoid space?

77. Where is CSF reabsorbed into the blood?

Hydrocephalus

78. Distinguish communicating from non-communicating (obstructive) hydrocephalus.

79. What is the classic clinical triad of normal-pressure hydrocephalus?

80. In one sentence, what is hydrocephalus?

Cranial Nerves

81. Which cranial nerve carries widespread parasympathetic fibres?

82. Which cranial nerve supplies the muscles of facial expression?

83. Which extraocular muscle does the trochlear nerve (CN IV) supply?

84. How many divisions does the trigeminal nerve (CN V) have, and name them.

Answer Key

Worked answers, grouped to match the questions above.

Fertilization & the Acrosome

1. The sperm would be unable to penetrate the zona pellucida. The acrosome stores hydrolytic enzymes (e.g. acrosin and hyaluronidase) that are released in the acrosome reaction to digest the zona pellucida's glycoprotein coat; without functional enzymes the sperm cannot breach the zona to reach the oocyte. Sperm motility and spermatogenesis are unaffected, as they do not depend on acrosomal enzymes.

Implantation & the Placenta

2. The trophoblast (its invasive cyto- and syncytiotrophoblast). The trophoblast first appears at the blastocyst stage (blastulation), so that is the earliest stage at which abnormal attachment to the uterine wall could begin.

The Female Reproductive System

3. The ovaries, the uterine (fallopian) tubes, the uterus, and the vagina — together with the external genitalia (vulva). It therefore consists of more than the ovaries, tubes, uterus and vagina alone.

4. About every 28 days (roughly once a month), not every 14 days. Ovulation falls near day 14, the mid-point of the cycle.

5. From puberty (about 11-14 years) to menopause (about 45-55 years).

6. In the endometrium of the uterus. The parallel ovarian cycle takes place in the ovary; the two are coordinated but are distinct cycles.

The Ovarian Cycle

7. It is described in three parts - the follicular phase, ovulation, and the luteal phase (often summarised as two major phases: follicular and luteal). It is not four phases.

8. At about the middle of the cycle - around day 14 of a 28-day cycle.

9. The follicular phase occupies the first half (before ovulation); the luteal phase occupies the second half (after ovulation).

10. The preovulatory phase, or the estrogenic phase - because rising estrogen dominates it.

The Sequence of Fertilization

11. (1) The sperm penetrates the corona radiata, then (2) acrosomal enzymes digest the zona pellucida, then (3) the sperm binds and fuses with the oolemma (vitelline membrane), and finally (4) the cortical reaction hardens the zona to block further sperm (polyspermy).

Gametogenesis (Oogenesis & Spermatogenesis)

12. 46 chromosomes and 92 chromatids. DNA has already been duplicated in S phase, so each of the 46 chromosomes consists of two sister chromatids (a 2n, 4c cell).

13. Oogenesis yields a single functional ovum plus 2-3 non-functional polar bodies; spermatogenesis yields four functional spermatozoa and no polar bodies.

14. Oogenesis - the ovum is far larger than the spermatozoon, because the egg retains almost all of the cytoplasm.

15. About two months (roughly 64-74 days).

Ovulation

16. The secondary oocyte (arrested in metaphase II) is expelled with the follicular fluid from the ovarian (Graafian) follicle. It is not released from the zona pellucida, which travels with the oocyte.

Sperm Production & Storage

17. They are produced in the seminiferous tubules of the testis and are stored (and further matured) in the epididymis.

The Preovulatory (Follicular) Phase

18. Secretion of estrogen. The phase therefore involves more than follicle maturation alone.

19. The developing (Graafian) follicle.

20. The proliferative phase - estrogen stimulates regrowth and thickening of the endometrium.

The Second Week of Development

21. Implantation of the blastocyst into the endometrium is completed.

22. It is a layer of small cuboidal cells on the ventral surface of the inner cell mass, lying adjacent to the blastocyst (blastocoel) cavity, appearing around day 6-7.

23. Vacuoles appear in the syncytiotrophoblast and coalesce into larger trophoblastic lacunae - the beginning of the uteroplacental circulation.

24. The exocoelomic (Heuser's) membrane together with the hypoblast.

The Embryonic Germ Layers

25. The ectoderm (neuroectoderm) - not the mesoderm.

26. The urogenital system - organs of the reproductive and excretory (urinary) systems - along with muscle, bone, blood, the heart and the kidneys.

27. The endoderm.

Hormonal Control of the Ovarian Cycle

28. Follicle-stimulating hormone (FSH).

29. Negative feedback - estrogen generally suppresses FSH release. The exception is the mid-cycle estrogen peak, which exerts positive feedback to trigger the LH surge.

30. Luteinizing hormone (LH).

Capacitation

31. It is the final functional maturation of sperm, occurring within the female reproductive tract. Surface glycoproteins and seminal-plasma proteins coating the acrosome are removed, priming the sperm to undergo the acrosome reaction.

32. The tail becomes hyperactivated (activated, not inactivated) - a vigorous whip-like motility that helps the sperm penetrate the egg's coverings.

33. Only capacitated sperm can penetrate the corona radiata and undergo the acrosome reaction needed to breach the zona pellucida.

The Zona Pellucida

34. It is the translucent coat surrounding the oocyte, composed mainly of glycoproteins (ZP1-ZP3) - not glycolipids. It lies immediately around the oocyte, not between the theca interna and externa.

35. ZP3 (zona glycoprotein 3); binding of the sperm to ZP3 triggers the acrosome reaction.

Organization of the Nervous System

36. The brain and the spinal cord.

37. The cranial and spinal nerves (and their ganglia).

38. Voluntary functions (e.g. skeletal-muscle movement). Involuntary visceral functions are handled by the autonomic nervous system.

39. The meninges.

Neuroglia

40. Specialised, non-neuronal supporting cells of the nervous system.

41. Astrocytes, through their perivascular end-feet.

42. The axon.

43. Ependymal cells.

The Hindbrain

44. The cerebellum, medulla oblongata and pons. The cerebrum is part of the forebrain, not the hindbrain.

Synaptic Transmission

45. From the presynaptic terminal (the axon terminal of the transmitting neuron).

46. Into the synaptic cleft, which is roughly 20-40 nm wide. They are not released into the axon terminal itself.

47. The dendrites.

The Autonomic Nervous System

48. The sympathetic division.

49. Involuntary (automatic). Voluntary actions such as running are somatic, not autonomic.

50. Norepinephrine - except the fibres to sweat glands (and a few others), which release acetylcholine.

51. Yes - it is considered a division of the autonomic nervous system.

The Cerebral Cortex

52. The parietal lobe.

53. The frontal lobe.

54. Vision is the occipital lobe; hearing is the temporal lobe. Smell is not an occipital-lobe function.

The Spinal Cord

55. The foramen magnum.

56. The intervertebral foramen.

57. 31 pairs (not 32).

58. At about L1-L2, where it tapers to the conus medullaris.

The Blood-Brain Barrier

59. Tight junctions between the cerebral capillary endothelial cells.

60. Pericytes and astrocytic end-feet.

61. No - it is relatively immature at birth and matures during early development.

62. Lipid-soluble substances - they diffuse through the lipid membranes down their concentration gradient.

Circumventricular Organs (CVOs)

63. They lack a full blood-brain barrier and are therefore highly permeable, letting the brain sample and secrete into the blood.

64. The area postrema.

65. Around the midline margins of the ventricles - not in the cerebral cortex.

Transport Across the BBB

66. By facilitated diffusion using the GLUT1 transporter.

67. By carrier-mediated (facilitated) transport.

68. No. Many are lipid-soluble, but some water-soluble drugs enter through specific carrier/transporter systems.

BBB Breakdown in Disease

69. Meningitis, stroke, multiple sclerosis (T-cell infiltration through a compromised barrier), and traumatic brain injury.

70. Harmful - it promotes vasogenic edema and worsens the injury; it is not part of tissue recovery.

71. Partly because disruption of the blood-brain barrier lets fluid and proteins leak into brain tissue.

Cerebrospinal Fluid (CSF) Production

72. By the choroid plexus within the ventricles.

73. About 500 mL is produced daily, while the total volume is only about 150 mL - so the fluid is turned over several times a day.

74. No - production stays relatively constant and is largely independent of intracranial pressure.

CSF Circulation

75. The interventricular foramen (foramen of Monro).

76. The foramina of Luschka (lateral) and the foramen of Magendie (median).

77. At the arachnoid granulations, which return CSF to the dural venous sinuses - not at the choroid plexus, which produces it.

Hydrocephalus

78. Non-communicating (obstructive) hydrocephalus results from a blockage within the ventricular system; communicating hydrocephalus results from impaired CSF reabsorption while flow between the ventricles is preserved.

79. Dementia, gait disturbance, and urinary incontinence.

80. An abnormal accumulation of cerebrospinal fluid within the ventricles of the brain.

Cranial Nerves

81. The vagus nerve (CN X).

82. The facial nerve (CN VII).

83. The superior oblique - not the superior rectus.

84. Three: the ophthalmic (V1), maxillary (V2) and mandibular (V3) divisions.

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